

RS232 Protocol Function List

Baud Rate : 19200
 Data Bits: 8
 Parity: None
 Stop Bits: 1
 Flow Control : None
 UART16550 FIFO: Disable

Note: It supports data rates from 9600 to 115200 bps. The default is 19200 bps. Depending on the equipment, a lower baud rate may be recommended for long cable runs.

Write Command

~	X	X	X	X	X		n	CR
Lead Code	Projector ID		Command			space	variable	carriage return
Prefix	00~99 (00: All projectors)		000~999				0~9999	suffix

Response Format

Pass: Fail:

Read Command

~	X	X	X	X	X		n	CR
Lead Code	Projector ID		Command			space	variable	carriage return
Prefix	00~99 (00: All projectors)		000~999				0~9999	suffix

Response Format

Pass:

O	k	n
		Variable

 Fail:

System Automatically Send

I	N	F	O	n
				variable

Note: There is a <CR> after all ASCII commands. 0D is the HEX code for <CR> in ASCII code.

Character to HEX table

Character	HEX
~	7E
0	30
1	31
2	32
3	33
4	34
5	35
6	36
7	37
8	38
9	39
Space	20
CR (Enter)	0D

e.g.
ASCII to HEX

RS232 Command	~	0	0	1	9	5		1	CR
HEX	7E	30	30	31	39	35	20	31	0D

Display Menu

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter			
Image Settings	Display Mode			[No Signal]						~XX123		1	F	O	k	0	
				Presentation	~XX20		1	F	P	~XX123		1	F	O	k	1	
				Bright	~XX20		2	F	P	~XX123		1	F	O	k	2	
				Cinema	~XX20		3	F	P	~XX123		1	F	O	k	3	
				HDR	~XX20		21	F	P	~XX123		1	F	O	k	21	
				sRGB	~XX20		4	F	P	~XX123		1	F	O	k	4	
				DICOM SIM.	~XX20		13	F	P	~XX123		1	F	O	k	10	
				Blending	~XX20		19	F	P	~XX123		1	F	O	k	19	
				User	~XX20		5	F	P	~XX123		1	F	O	k	5	
			3D	~XX20		9	F	P	~XX123		1	F	O	k	9		
	Wall Color			Off	~XX506		0	F	P								
				Blackboard	~XX506		1	F	P								
				Light Yellow	~XX506		7	F	P								
				Light Green	~XX506		3	F	P								
				Light Blue	~XX506		4	F	P								
				Pink	~XX506		5	F	P								
				Gray	~XX506		6	F	P								
	Dynamic Range	HDR		Off	~XX565		0	F	P								
				Auto	~XX565		1	F	P								
		HDR Picture Mode		Bright	~XX566		0	F	P								
				Standard	~XX566		1	F	P								
				Film	~XX566		2	F	P								
				Detail	~XX566		3	F	P								
				SMPTE 2084	~XX566		4	F	P								
	Brightness			-	~XX46		1	F	P								
				0~100	~XX21		0~100	F	P	~XX125		1	F	O	k	0~100	
				+	~XX46		2	F	P								
	Contrast			-	~XX47		1	F	P								
				0~100	~XX22		0~100	F	P	~XX126		1	F	O	k	0~100	
				+	~XX47		2	F	P								
	Sharpness			1 ~ 15	~XX23		1~15	F	P								
	Color			0~100	~XX45		0~100	F	P								
	Tint			0~100	~XX44		0~100	F	P								

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter		
Image Settings	Gamma	Film			~XX35		1	F	P							
		Video			~XX35		2	F	P							
		Graphics			~XX35		3	F	P							
		Standard(2.2)			~XX35		4	F	P							
		3D														
		Blackboard														
		DICOM SIM.			~XX35		11	F	P							
		1.8			~XX35		5	F	P							
		2.0			~XX35		6	F	P							
		2.4			~XX35		12	F	P							
	Color Settings	BrilliantColor™		0~10	~XX34		0~10	F	P							
		Color Temperature		Warm	~XX36		4	F	P	~XX128		1	F	O	k	3
				Standard	~XX36		1	F	P	~XX128		1	F	O	k	0
				Cool	~XX36		2	F	P	~XX128		1	F	O	k	1
		Color Matching	Color	R / G / B / C / Y / M / W												
			(R) Hue	0~254	~XX327		0~254	F	P							
			(R) Saturation	0~254	~XX333		0~254	F	P							
			(R) Gain	0~254	~XX339		0~254	F	P							
			(G) Hue	0~254	~XX328		0~254	F	P							
			(G) Saturation	0~254	~XX334		0~254	F	P							
			(G) Gain	0~254	~XX340		0~254	F	P							
			(B) Hue	0~254	~XX329		0~254	F	P							
			(B) Saturation	0~254	~XX335		0~254	F	P							
			(B) Gain	0~254	~XX341		0~254	F	P							
			(C) Hue	0~254	~XX330		0~254	F	P							
			(C) Saturation	0~254	~XX336		0~254	F	P							
			(C) Gain	0~254	~XX342		0~254	F	P							
			(M) Hue	0~254	~XX332		0~254	F	P							
			(M) Saturation	0~254	~XX338		0~254	F	P							
			(M) Gain	0~254	~XX344		0~254	F	P							
			(Y) Hue	0~254	~XX331		0~254	F	P							
			(Y) Saturation	0~254	~XX337		0~254	F	P							
			(Y) Gain	0~254	~XX343		0~254	F	P							
			(W) Red	0~254	~XX345		0~254	F	P							
			(W) Green	0~254	~XX346		0~254	F	P							
			(W) Blue	0~254	~XX347		0~254	F	P							

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter
Image Settings	Color Settings	Color Matching	Reset	Cancel / Yes	~XX215	1	F	P						
			Exit											
		RGB Gain/Bias	Red Gain	0~100	~XX24	0~100	F	P						
			Green Gain	0~100	~XX25	0~100	F	P						
			Blue Gain	0~100	~XX26	0~100	F	P						
			Red Bias	0~100	~XX27	0~100	F	P						
			Green Bias	0~100	~XX28	0~100	F	P						
			Blue Bias	0~100	~XX29	0~100	F	P						
			Reset	Cancel / Yes	~XX517	1	F	P						
			Exit											
		Color Space		Auto	~XX37	1	F	P						
				RGB	~XX37	2	F	P						
				YUV	~XX37	3	F	P						
				RGB(0~255)	~XX37	2	F	P						
				RGB(16~235)	~XX37	4	F	P						
	UltraDetail			Off	~XX41	0	F	P						
				1	~XX41	4	F	P						
				2	~XX41	5	F	P						
				3	~XX41	6	F	P						
	Extreme Black			Off	~XX218	0	F	P						
				On	~XX218	1	F	P						
	Dynamic Black			Off	~XX191	0	F	P						
				On	~XX191	1	F	P						
	Brightness Mode	Constant Power			~XX110	6	F	P						
		Constant Luminance			~XX110	7	F	P						
		Eco Mode			~XX110	2	F	P						
	Power Level			1 ~ 100	~XX326	1~100	F	P						
	Pure Engine	PureContrast		Off	~XX219	0	F	P						
				On	~XX219	1	F	P						
		PureColor		Off	~XX42	0	F	P						
				1	~XX42	1	F	P						
				2	~XX42	2	F	P						
				3	~XX42	3	F	P						
				4	~XX42	4	F	P						
				5	~XX42	5	F	P						

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter		
Image Settings	Pure Engine	PureMotion		Off	~XX190	0	F	P								
				1	~XX190	1	F	P								
				2	~XX190	2	F	P								
				3	~XX190	3	F	P								
		PureMotion Demo		Off	~XX197	0	F	P								
				H Split	~XX197	1	F	P								
				V Split	~XX197	2	F	P								
	Reset				~XX509	1	F	P								
3D	3D Mode			On	~XX230	4	F	P								
				Off	~XX230	0	F	P								
	3D Format			Auto	~XX405	0	F	P								
				Frame Packing	~XX405	7	F	P								
				Side by Side	~XX405	1	F	P								
				Top and Bottom	~XX405	2	F	P								
				Frame Sequential	~XX405	3	F	P								
	3D-2D			3D	~XX400	0	F	P								
				L	~XX400	1	F	P								
				R	~XX400	2	F	P								
	3D Sync Invert			Off	~XX231	0	F	P								
				On	~XX231	1	F	P								
	3D Sync Out			To Emitter	~XX232	0	F	P								
				To Next Projector	~XX232	1	F	P								
	L/R Reference			Field GPIO	~XX236	0	F	P								
				1ST Frame	~XX236	1	F	P								
	3D Frame Delay			1~200	~XX233	1~200	F	P								
	Reset				~XX234	1	F	P								
Aspect Ratio				4:3	~XX60	1	F	P	~XX127	1	F	O	k	1		
				16:9	~XX60	2	F	P	~XX127	1	F	O	k	2		
				16:10	~XX60	3	F	P	~XX127	1	F	O	k	3		
				LBX	~XX60	5	F	P	~XX127	1	F	O	k	5		
				Native	~XX60	6	F	P	~XX127	1	F	O	k	6		
				Auto	~XX60	7	F	P	~XX127	1	F	O	k	7		
Digital Zoom	H Zoom			50 ~ 400 (%)	~XX504	50~400	F	P	~XX543	8	F	O	k	50~400		
	V Zoom			50 ~ 400 (%)	~XX505	50~400	F	P	~XX543	7	F	O	k	50~400		
	Reset															
	Exit															

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter		
Image shift	(icon)H			0~100	~XX63		0~100	F	P	~XX543		1	F	O	k	0~100
	(icon)V			0~100	~XX64		0~100	F	P	~XX543		2	F	O	k	0~100
	H Arc			0~100	~XX300		0~100	F	P	~XX543		6	F	O	k	0~100
Geometric Correction	V Arc			0~100	~XX301		0~100	F	P	~XX543		5	F	O	k	0~100
	Four Corners			Top-left	~XX581		0~120	F	P							
					~XX582		0~80	F	P							
					~XX59		1	F	P							
					~XX59		2	F	P							
					~XX59		3	F	P							
					~XX59		4	F	P							
				Top-right	~XX583		0~120	F	P							
					~XX584		0~80	F	P							
					~XX59		5	F	P							
					~XX59		6	F	P							
					~XX59		7	F	P							
					~XX59		8	F	P							
				Bottom-left	~XX585		0~120	F	P							
					~XX586		0~80	F	P							
					~XX59		9	F	P							
					~XX59		10	F	P							
					~XX59		11	F	P							
					~XX59		12	F	P							
				Bottom-right	~XX587		0~120	F	P							
					~XX588		0~80	F	P							
					~XX59		13	F	P							
					~XX59		14	F	P							
					~XX59		15	F	P							
					~XX59		16	F	P							
	H Keystone			0~40	~XX65		0~40	F	P	~XX543		4	F	O	k	0~40
	V Keystone			0~40	~XX66		0~40	F	P	~XX543		3	F	O	k	0~40
	Warp and Blend Control	PC Connection		Yes						~XX132		3	F	O	k	1
				No					~XX132		3	F	O	k	0	
	Warp and Blend Control	Settings		All Off	~XX142		0	F	P	~XX132		1	F	O	k	0
				All On	~XX142		3	F	P	~XX132		1	F	O	k	3
				Blend Off	~XX142		4	F	P	~XX132		1	F	O	k	4

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Level 4	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter		
Geometric Correction	Warp and Blend Control	Memory		Off	~XX147	0	F	P	~XX137	1	F	O	k	0		
				User1	~XX147	1	F	P	~XX137	1	F	O	k	1		
				User2	~XX147	2	F	P	~XX137	1	F	O	k	2		
				User3	~XX147	3	F	P	~XX137	1	F	O	k	3		
	Reset				~XX561	1	F	P								
PIP/PBP	Screen			Off	~XX302	0	F	P								
				PIP	~XX302	1	F	P								
				PBP	~XX302	2	F	P								
	Location			PIP-Bottom Right	~XX303	4	F	P								
				PIP-Bottom Left	~XX303	3	F	P								
				PIP-Top Left	~XX303	1	F	P								
				PIP-Top Right	~XX303	2	F	P								
				PBP, Main Left	~XX303	5	F	P								
				PBP, Main Top	~XX303	6	F	P								
				PBP, Main Right	~XX303	7	F	P								
				PBP, Main Bottom	~XX303	8	F	P								
	Size			Small	~XX304	3	F	P								
				Medium	~XX304	2	F	P								
				Large	~XX304	1	F	P								
	Main Source	HDMI1			~XX12	1	F	P	~XX121	1	F	O	k	7		
		HDMI2			~XX12	15	F	P	~XX121	1	F	O	k	8		
		DisplayPort			~XX12	20	F	P	~XX121	1	F	O	k	15		
		HDBaseT			~XX12	21	F	P	~XX121	1	F	O	k	16		
		3G-SDI			~XX12	22	F	P	~XX121	1	F	O	k	18		
	Sub Source	HDMI1			~XX305	1	F	P	~XX131	1	F	O	k	7		
		HDMI2			~XX305	4	F	P	~XX131	1	F	O	k	8		
		DisplayPort			~XX305	17	F	P	~XX131	1	F	O	k	15		
		HDBaseT			~XX305	10	F	P	~XX131	1	F	O	k	16		
		3G-SDI			~XX305	11	F	P	~XX131	1	F	O	k	17		
	Swap				~XX306	1	F	P								

Note: Some commands are not supported depending on the model.

Setup Menu

Submenu Level 1	Level 2	Level 3	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter			
Projection			Front	~XX71		1	F	P	~XX129		1	F	O	k	0	
			Rear	~XX71		2	F	P	~XX129		1	F	O	k	1	
			Ceiling-top	~XX71		3	F	P	~XX129		1	F	O	k	2	
			Rear-top	~XX71		4	F	P	~XX129		1	F	O	k	3	
Lens Settings	Zoom		+	~XX307		1	F	P								
			-	~XX307		2	F	P								
	Focus		+	~XX308		1	F	P								
			-	~XX308		2	F	P								
	Lens Function		Lock	~XX349		1	F	P	~XX545		4	F	O	k	0	
			Unlock	~XX349		2	F	P	~XX545		4	F	O	k	1	
	Lens Shift		Up	~XX84		3	F	P								
			Down	~XX84		4	F	P								
			Left	~XX84		5	F	P								
			Right	~XX84		6	F	P								
	Lens Calibration		Yes	~XX525		1	F	P								
			NO	~XX525		0	F	P								
	Lens Memory	Apply Position	1~5	~XX359		1~5	F	P								
		Save Current Position	1~5	~XX360		1~5	F	P								
Reset		Yes/No	~XX361		1	F	P									
Power Settings	Direct Power On		Off	~XX105		0	F	P								
			On	~XX105		1	F	P								
	Auto Power Off (min.)		0 ~ 180 (5 min increments)	~XX106		0~180	F	P								
	Sleep Timer (min.)	Sleep Timer (min.)	0 ~ 990 (30 min increments)	~XX107		000~990	F	P								
		Always On	No	~XX507		0	F	P								
			Yes	~XX507		1	F	P								
	Power Mode (Standby)		Eco	~XX114		0	F	P	~XX150		16	F	O	k	1	
			Active	~XX114		1	F	P	~XX150		16	F	O	k	0	
		Communications	~XX114		3	F	P	~XX150		16	F	O	k	3		
Security	Security		Off	~XX78		0 ~nnnn	F	P								
			On	~XX78		1 ~nnnn	F	P								
	Security Timer	Month		~XX537		00~12	F	P	~XX544		1	F	O	k	00~12	
		Day		~XX538		00~29	F	P	~XX544		2	F	O	k	00~29	
		Hour		~XX539		00~23	F	P	~XX544		3	F	O	k	00~23	
		MM/DD/HH (RS232 only)		~XX77		MMDDHH	F	P								

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter			
Security	Change Password															
Test Pattern			Off	~XX195	0	F	P									
			Green Grid	~XX195	3	F	P									
			Magenta Grid	~XX195	4	F	P									
			White Grid	~XX195	1	F	P									
			White	~XX195	2	F	P									
			Black	~XX195	11	F	P									
			Red	~XX195	5	F	P									
			Green	~XX195	6	F	P									
			Blue	~XX195	7	F	P									
			Yellow	~XX195	8	F	P									
			Magenta	~XX195	9	F	P									
			Cyan	~XX195	10	F	P									
Remote Settings [depends on remote]	IR Function	Front	Off	~XX11	4	F	P	~XX542	1	F	O	k	0			
			On	~XX11	5	F	P	~XX542	1	F	O	k	1			
		Top	Off	~XX11	6	F	P	~XX542	2	F	O	k	0			
			On	~XX11	7	F	P	~XX542	2	F	O	k	1			
		HDBaseT	Off	~XX11	10											
			On	~XX11	9											
	Remote Code		0~99	~XX350	00~99	F	P					O	k	00~99		
	Hot-Key Settings		Aspect Ratio/Freeze Screen	~XX117	1~2	F	P									
12V Trigger			Off	~XX192	0	F	P									
			On	~XX192	1	F	P									
ProService																
Projector ID			0~99	~XX79	00~99	F	P	~XX558	1	F	O	k	00~99			
Light Sensor			Default	~XX552	0	F	P									
			Manual	~XX552	2	F	P									
Keypad LED Settings			Off	~XX362	0	F	P									
			On	~XX362	1	F	P									
Options	Language		English	~XX70	1	F	P									
			Deutsch	~XX70	2	F	P									
			Français	~XX70	3	F	P									
			Italiano	~XX70	4	F	P									
			Español	~XX70	5	F	P									
			Português	~XX70	6	F	P									

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter		
Options	Language		Polski	~XX70	7	F	P								
			Nederlands	~XX70	8	F	P								
			Svenska	~XX70	9	F	P								
			Norsk/Dansk	~XX70	10	F	P								
			Suomi	~XX70	11	F	P								
			ελληνικά	~XX70	12	F	P								
			繁體中文	~XX70	13	F	P								
			簡体中文	~XX70	14	F	P								
			日本語	~XX70	15	F	P								
			한국어	~XX70	16	F	P								
			Русский	~XX70	17	F	P								
			Magyar	~XX70	18	F	P								
			Čeština	~XX70	19	F	P								
			ไทย	~XX70	21	F	P								
			Türkçe	~XX70	22	F	P								
			Tiếng Việt	~XX70	25	F	P								
			Bahasa Indonesia	~XX70	26	F	P								
			Română	~XX70	27	F	P								
			Slovakian	~XX70	28	F	P								
	Menu Settings	Menu Location	Top left	~XX72	1	F	P								
			Top right	~XX72	2	F	P								
			Center	~XX72	3	F	P								
			Bottom left	~XX72	4	F	P								
			Bottom right	~XX72	5	F	P								
		Menu Transparency	0~9	~XX526	0~9	F	P								
		Menu Timer	Off	~XX515	0	F	P								
			5sec	~XX515	1	F	P								
			10sec	~XX515	3	F	P								
			15Sec	~XX515	4	F	P								
	Auto Source		Off	~XX563	0	F	P								
			On	~XX563	1	F	P								
	Input Source		HDMI1	~XX12	1	F	P	~XX121	1	F	O	k	7		
			HDMI2	~XX12	15	F	P	~XX121	1	F	O	k	8		
			DisplayPort	~XX12	20	F	P	~XX121	1	F	O	k	15		
			HDBaseT	~XX12	21	F	P	~XX121	1	F	O	k	16		
			3G-SDI	~XX12	22	F	P	~XX121	1	F	O	k	18		

Note: Some commands are not supported depending on the model.

Submenu Level 1	Level 2	Level 3	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter			
Options	High Altitude		Off	~XX101		0	F	P								
			On	~XX101		1	F	P								
	Information Hide		Off	~XX102		0	F	P								
			On	~XX102		1	F	P								
	Logo		Default	~XX82		1	F	P								
			Neutral	~XX82		3	F	P								
	Background Color		None	~XX104		0	F	P								
			Blue	~XX104		1	F	P								
			Red	~XX104		3	F	P								
			Green	~XX104		4	F	P								
			Gray	~XX104		6	F	P								
			Logo	~XX104		7	F	P								
	Serial Port Baud Rate		9600							~XX153		1	F	O	k	9600
			14400							~XX153		1	F	O	k	14400
			19200							~XX153		1	F	O	k	19200
			38400							~XX153		1	F	O	k	38400
			57600							~XX153		1	F	O	k	57600
			115200							~XX153		1	F	O	k	115200
	Serial Port Path		RS232	~XX557		1	F	P								
			HDBaseT	~XX557		2	F	P								
	EDID Reminder		Off													
			On													
	HDMI 1 EDID		1.4													
			2.0													
	HDMI 2 EDID		1.4													
			2.0													
	HDBaseT EDID		1.4													
			2.0													
System Update	Notification		On	~XX168		1	F	P	~XX158		1	F	O	k	1	
			Off	~XX168		0	F	P	~XX158		1	F	O	k	0	
	System Update		Cancel / Yes	~XX168		9	F	P								
Reset	Reset OSD		Cancel / Yes	~XX546		1	F	P								
	Reset to Default		Cancel / Yes	~XX112		1	F	P								

Note: Some commands are not supported depending on the model.

Network Menu

Submenu Level 1	Level 2	Level 3	Value	Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter
WLAN	WLAN		Off	~XX450	0	F	P						
			On	~XX450	1	F	P						
	Subnet Mask		255.255.255.0										
	IP Address		192.168.1.1						~XX451	2	F	O	k nnn.nnn.nnn.nnn
	Start IP								~XX451	5	F	O	k nnn.nnn.nnn.nnn
	End IP								~XX451	6	F	O	k nnn.nnn.nnn.nnn
	SSID		(Read only) Network Display						~XX451	3	F	O	k nnnnnnnnnnnnnnn
	Apply												
LAN	MAC Address		(Read only)						~XX555	1	F	O	k nn:nn:nn:nn:nn
	DHCP		Off						~XX150	17	F	O	k 0
			On						~XX150	17	F	O	k 1
	IP Address		192.168.0.100						~XX87	3	F	O	k nnn.nnn.nnn.nnn
	Subnet Mask		255.255.255.0										
	Gateway		192.168.0.51										
	DNS		0.0.0.0										
	Apply												
Control	Crestron		Off	~XX454	0	F	P						
			On	~XX454	1	F	P						
	Extron		Off	~XX455	0	F	P						
			On	~XX455	1	F	P						
	PJ Link		Off	~XX456	0	F	P						
			On	~XX456	1	F	P						
	AMX Device Discovery		Off	~XX457	0	F	P						
			On	~XX457	1	F	P						
	Telnet		Off	~XX458	0	F	P						
			On	~XX458	1	F	P						
Network Reset			Off	~XX459	0	F	P						
			On	~XX459	1	F	P						

Note: Some commands are not supported depending on the model.

Information Menu

Submenu Level 1	Value	Read Command	Space	Command Value	Fail	Reply Parameter		
Regulatory								
Serial Number		~XX353	1		F O k	nnnnnnnnnnnnnnnn		
Main Source		~XX121	1		F	* Check "Input Source" read above		
Main Resolution		~XX150	4		F O k	nnnnnnnnnnnn (e.g. Ok1920x1080)		
Main Signal Format		~XX150	5		F O k	nnnnnnnnnnnn (e.g. OkDVI-D)		
Main Pixel Clock		~XX150	6		F O k	nnnnnnnnnnnn		
Main Horz Refresh		~XX150	7		F O k	nnnnnnnnnnnn (e.g. Ok00x00)		
Main Vert Refresh		~XX150	8		F O k	nnnnn (e.g. Ok60Hz)		
Sub Source		~XX131	1		F	* Check "Input Source" read above		
Sub Resolution		~XX150	10		F O k	nnnnnnnnnnnn (e.g. Ok1920x1080)		
Sub Signal Format		~XX150	11		F O k	nnnnnnnnnnnn (e.g. OkDVI-D)		
Sub Pixel Clock		~XX150	12		F O k			
Sub Horz Refresh	00x00	~XX150	13		F O k	nnnnnnnnnnnn (e.g. Ok00x00)		
Sub Vert Refresh	0.00Hz	~XX545	14		F O k	nnnnn (e.g. Ok60Hz)		
Display Mode		~XX123	1		F	* Check "Input Source" read above		
System Temperature		~XX150	18		F O k	nnn= string (e.g. Ok48)		
Color Depth		~XX156	1		F O k	nbit nnn (e.g. 8bit RGB)		
Color Format		~XX157	1		F O k	nnnnnnnnnnnn (eg. BT.2020 HDR)		
Power Mode (Standby)	Eco	~XX150	16		F O k	0		
	Active	~XX150	16		F O k	1		
	Communications	~XX150	16		F O k	3		
Projection Hours		~XX108	1		F O k	nnnnn (nnnnn= hour digits)		
Remote Code	0~99							
LAN IP Address		~XX87	3		F O k	nnn.nnn.nnn.nnn		
SSID		~XX451	3		F O k	nnnnnnnnnnnnnnnn		
Projector ID	0~99	~XX558	1		F O k	nn		
Brightness Mode								
Power Level	1~100							
FW Version	Global version	~XX122	1		F O k	nnnnnnnnnnnnnnnn		
F-MCU Version								
S-MCU Version								
F-Image Version								
Formatter Version								
LAN Version								

Note: Some commands are not supported depending on the model.

Other Uart Command

Command			Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter			
Power Off			~XX00		0 / 2	F	P	~XX124		1	F		O	k	0
Power On			~XX00		1	F	P	~XX124		1	F		O	k	1
Re-Sync			~XX01		1	F	P								
AV Mute	Off		~XX02		0	F	P	~XX355		1	F		O	k	0
	On		~XX02		1	F	P	~XX355		1	F		O	k	1
Freeze	Unfreeze		~XX04		0	F	P								
	Freeze		~XX04		1	F	P								
3D Sync Out	To Emitter		~XX232		0										
	To Next Projector		~XX232		1										
3D Frame Delay	1~200		~XX233		1~nnn										
Output 3D state	2D							~XX130		1	F		O	k	0
	3D							~XX130		1	F		O	k	1
System Auto Send	Standby Mode											I	N	F	O 0
	Warming up											I	N	F	O 1
	Cooling Down											I	N	F	O 2
	Out of Range											I	N	F	O 3
	Lamp Fail (LED Fail)											I	N	F	O 4
	Thermal Switch Error											I	N	F	O 5
	Fan Lock											I	N	F	O 6
	Over Temperature											I	N	F	O 7
	Lamp Hours Running Out											I	N	F	O 8
	Cover Open											I	N	F	O 9
	Lamp Ignite Fail											I	N	F	O 10
	Format Board Power On Fail											I	N	F	O 11
	Color Wheel Unexpected Stop											I	N	F	O 12
	Over Temperature											I	N	F	O 13
	FAN 1 Lock											I	N	F	O 14
	FAN 2 Lock											I	N	F	O 15
	FAN 3 Lock											I	N	F	O 16

Note: Some commands are not supported depending on the model.

Command			Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter				
System Auto Send	FAN 4 Lock										I	N	F	O	17	
	FAN 5 Lock										I	N	F	O	18	
	LAN fail then restart										I	N	F	O	19	
	LD lower than 60%										I	N	F	O	20	
	LD NTC (1) Over Temperature										I	N	F	O	21	
	LD NTC (2) Over Temperature										I	N	F	O	22	
	High Ambient Temperature										I	N	F	O	23	
	System Ready										I	N	F	O	24	
Model Name	Optoma WUXGA						~XX151	1	F			O	k	5		
	Optoma UHD						~XX151	1	F			O	k	6		
Software Version							~XX122	1	F			O	k	nnnnnnnn (Software Version)		
LAN FW version							~XX357	1	F			O	k	nnnnnnnn (LAN FW version)		
Fan Speed	Fan 1 Speed	0000~9999					~XX351	1	F			O	k	0000~9999		
	Fan 2 Speed	0000~9999					~XX351	2	F			O	k	0000~9999		
	Fan 3 Speed	0000~9999					~XX351	3	F			O	k	0000~9999		
	Fan 4 Speed	0000~9999					~XX351	4	F			O	k	0000~9999		
System Temperature							~XX352	1	F			O	k	0000~9999		
Information	Info String						~XX150	1	F			O	k	abbbbbcdddee (Note*1)		
	Native Resolution						~XX150	2	F			O	k	nnn..nn (e.g. Ok1920x1080)		
	Main Source						~XX150	3	F			O	k	nnn..nn (e.g. OkHDMI)		
	- Resolution						~XX150	4	F			O	k	nnn..nn (e.g.Ok1920x1080)		
	- Signal Format						~XX150	5	F			O	k	nnn..nn		
	- Pixel Clock						~XX150	6	F			O	k	nnn..nn		
	- Horz Refresh						~XX150	7	F			O	k	nnn..nn		
	- Vert Refresh						~XX150	8	F			O	k	nnn..nn		
	Sub Source						~XX150	9	F			O	k	nnn..nn		
	- Resolution						~XX150	10	F			O	k	nnn..nn (e.g. Ok1920x1080)		
	- Signal Format						~XX150	11	F			O	k	nnn..nn (e.g. OkHDMI)		
	- Pixel Clock						~XX150	12	F			O	k	nnn..nn		
	- Horz Refresh						~XX150	13	F			O	k	nnn..nn		
	- Vert Refresh						~XX150	14	F			O	k	nnn..nn		

Note: Some commands are not supported depending on the model.

Command			Write Command	Space	Set Parameter	Fail	Pass	Read Command	Space	Command Value	Fail	Reply Parameter				
Information	Light Source Mode							~XX150		15	F			O	k	nnn..nn
	Standby Power Mode	Active						~XX150		16	F			O	k	1
		Eco.						~XX150		16	F			O	k	2
	DHCP	Off						~XX150		17	F			O	k	0
		On						~XX150		17	F			O	k	1
	System Temperature							~XX150		18	F			O	k	nnn..nn (e.g. Ok48)
Serial Port Path	RS232		~XX557		1	F	P									
	HDBaseT		~XX557		2	F	P									
Source Lock	Off		~XX100		0	F	P									
	On		~XX100		1	F	P									
Filter Wheel Index			~XX528		0000~9999	F	P	~XX530		1	F			O	k	0000~9999
Phosphor Wheel Index			~XX529		0000~9999	F	P	~XX531		1	F			O	k	0000~9999
Light Sensor Calibration			~XX552		2	F	P									

Note 1																
Freeze Screen	Freeze can be release by menu key, exit key, and direct source key.															

Note: Some commands are not supported depending on the model.

Remote Control Simulation

Command	Write Command	Space	Set Parameter	Fail	Pass
Power	~XX140		1	F	P
Power Off	~XX140		2	F	P
Up	~XX140		10	F	P
Left	~XX140		11	F	P
Enter (for projection MENU)	~XX140		12	F	P
Right	~XX140		13	F	P
Down	~XX140		14	F	P
V Keystone +	~XX140		15	F	P
V Keystone -	~XX140		16	F	P
Brightness	~XX140		19	F	P
Menu	~XX140		20	F	P
Zoom	~XX140		21	F	P
AV Mute	~XX140		24	F	P
Contrast	~XX140		28	F	P
Lens shift	~XX140		31	F	P
Zoom +	~XX140		32	F	P
Zoom -	~XX140		33	F	P
Focus +	~XX140		34	F	P
Focus -	~XX140		35	F	P
Mode	~XX140		36	F	P
info	~XX140		40	F	P
Auto(Re-sync)	~XX140		41	F	P
Input(Source)	~XX140		47	F	P
1	~XX140		51	F	P
2	~XX140		52	F	P
3	~XX140		53	F	P
4	~XX140		54	F	P
5	~XX140		55	F	P
6	~XX140		56	F	P
7	~XX140		57	F	P
8	~XX140		58	F	P
9	~XX140		59	F	P
0	~XX140		60	F	P

Command	Write Command	Space	Set Parameter	Fail	Pass
Gamma	~XX140		61	F	P
PIP	~XX140		63	F	P
Lens H(left)	~XX140		64	F	P
Lens H(Right)	~XX140		65	F	P
Lens V(Up)	~XX140		66	F	P
Lens V(Down)	~XX140		67	F	P
H Keystone +	~XX140		68	F	P
H Keystone -	~XX140		69	F	P
Hot Key (user1)(F1)	~XX140		70	F	P
Pattern	~XX140		73	F	P
Exit	~XX140		74	F	P

Note: Some commands are not supported depending on the model.

